

Expressions/Grammar/Design Choices Relative to Lox:

My project currently follows most of Lox's rules and Grammar. I'm still looking into how I want to make my language more unique but I'm trying to follow Lox while I learn how to make AST printers.

The expression grammar of the project is as follows:

Expression →

- Literal
- Unary
- Binary
- Grouping

Literal →

- NUMBER
- STRING
- "True"
- "False"
- "None"

Unary →

- ("!" | "-") expression

Binary →

- Expression operator expression

Operator →

- ==
- !=
- <
- >
- >=
- <=
- +
- -
- *
- /

Grouping →

- "(" expression ")"

File/Test Instructions:

The lab2-AST_Tests.py file needs to be in the same folder / directory as the [main.py](#) file. When run, it produces a number of hard coded tests that cover a variety of combinations of expressions.

Test File Results:

When running the test file, there are 7 test cases. 4 of these cases test the individual expression categories while the rest do a combination of them all.

Literal Expression Test is meant to set a literal as 42 and print it to the terminal as that value. The result was 42, making this successful.

Binary Expression Test sets a binary expression of a literal 1 + a literal 2. This should output (1 + 2), which was successful.

Grouping Expression Test takes a grouping of a single literal 3, resulting in (3)

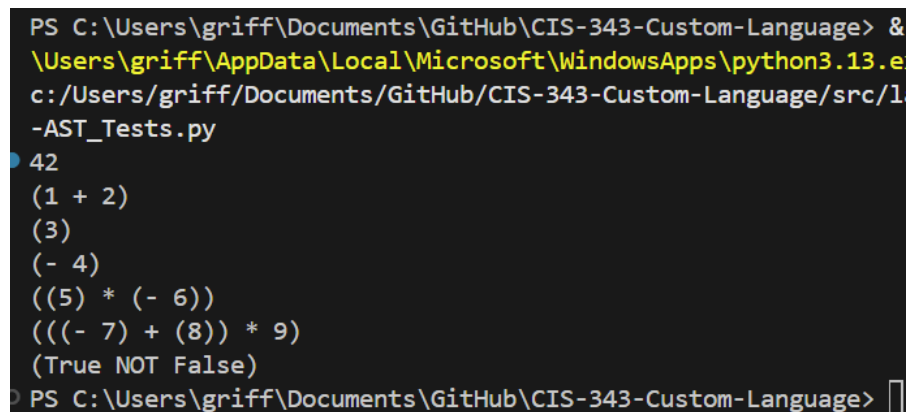
Unary Expression Test simply takes a - with literal 4 giving the output of (- 4)

Combo Test 1 takes Binary(Grouping(Literal(5)), "*", Unary("-", Literal(6))) and prints ((5) * (-6))

Combo Test 2 takes Binary(Binary(Unary("-", Literal(7)), "+", Grouping(Literal(8))), "*", Literal(9)) and prints (((-7)+(8))*9)

The final test is a quick combo test involving true and false values: Binary(Literal(True), "NOT", Literal(False)). This results in (True NOT False).

Image of Test Results:



```
PS C:\Users\griff\Documents\GitHub\CIS-343-Custom-Language> &
\Users\griff\AppData\Local\Microsoft\WindowsApps\python3.13.e
c:/Users/griff/Documents/GitHub/CIS-343-Custom-Language/src/1
-AST_Tests.py
42
(1 + 2)
(3)
(- 4)
((5) * (- 6))
((( - 7) + (8)) * 9)
(True NOT False)
PS C:\Users\griff\Documents\GitHub\CIS-343-Custom-Language>
```